



A computational model for stereolithography apparatus (SLA) 3D printing

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Abstract

Stereolithography apparatus (SLA) 3D printing involves scanning a laser across a photosensitive resin to cure it and build a part layer-by-layer. Conventionally, SLA printers are calibrated by conducting numerous experiments and analyzing the built parts to determine the optimal parameters that maximize the part quality while keeping the build-time low. This work aims to develop a computational model of the printer that characterizes the effect of these material and process parameters on the printing speed and part quality. This model could significantly reduce the time and capital required to calibrate these printers. The printing process is modeled by independently studying the resin curing and recoating processes and then combining the two phenomena. These studies were used to draw various insights into the printing process. A parametric analysis of the cure depth concluded that it significantly decays with increased laser scanning speed, whereas an optimum intermediate value of photoinitiator concentration maximizes it. A computational fluid dynamics (CFD) model of a gravity-based resin recoating process was developed and implemented in OpenFOAM®. Studies conducted using this model indicated that lower resin viscosity and higher surface tension lead to lower settling times. Inferences gained by studying the coupled model were used to develop a method to compensate for disturbances in the surface profile of the resin. SLA printer manufacturers could use the insights gained from these studies to optimize the printers to improve their speeds and print quality.

Keywords Vat photopolymerization · Stereolithography · Cure model · Resin flow · Computational fluid dynamics

1 Introduction

Stereolithography (SLA) 3D printing is an additive manufacturing process where a short wavelength laser is used to strike a photosensitive resin which causes chemical

monomers and oligomers to cross-link together to form polymers. This extensive cross-linking is also called curing, resulting in solidifying the selectively polymerized portion of the resin. This selectively polymerized resin is referred to as cured resin. A schematic of the SLA process is shown in Fig. 1. The foundational building block of parts manufactured using SLA are the “strings” of cured resin that form when a laser is scanned across a vat of photosensitive resin. Following the curing process, an SLA printer proceeds with the resin recoating process to prepare the vat for laser-scanning the next layer. This recoating mechanism varies considerably among SLA printers, especially among different manufacturers. Most printers use an elevator and a recoating blade for this purpose [1]. Printers must have precisely chosen elevator and blade motion parameters to ensure a feasible printing speed. It is crucial for printer manufacturers to optimize these parameters to ensure low build times. SLA printers are calibrated by conducting numerous experiments and analyzing the built parts to determine the optimal parameters that maximize the part quality while keeping the build-time low. However, such experiments require a lot of

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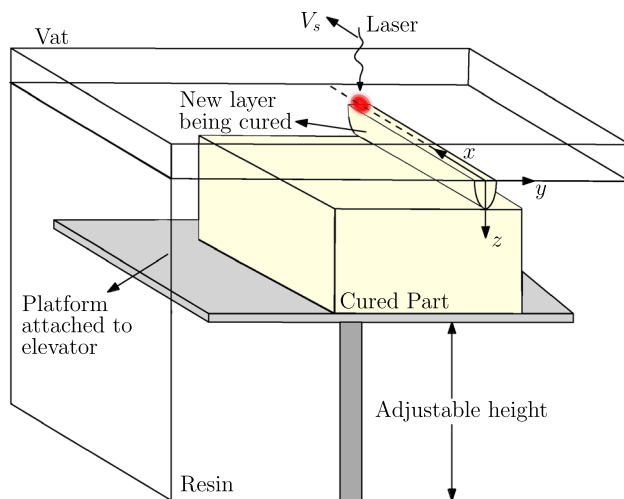


Fig. 1 Schematic of the SLA process

time and capital to be invested in them. Hence, there is a need for a model that can be used to identify appropriate process parameters for optimal part quality and build time. To that end, many researchers in academia and industry have tried to establish models that can accurately describe the physical phenomena involved in the curing and recoating processes of SLA 3D printers.

The first step to creating a model that can explain the underlying physics in SLA should be to model the process of photopolymerization of the resin using a laser. Jacobs [2] developed an energy exposure threshold model in 1993, and it was the first model developed to predict the cure profile of the SLA process and is still one of the most popular models in use today. It introduced the concept of a gel-point which defines the resin curing based on specific threshold energy. It introduced the working curve, a linear plot between the cure depth and the natural logarithm of the maximum actinic exposure energy. The model also predicted the generated cure profile to be of a parabolic cross-section. Despite the popularity of the energy exposure threshold model, it is seen that in many cases, the semi-logarithmic working curve plot is linear only for a specific range of energy exposures and deviates significantly for wide ranges of energy exposures [3, 4]. Moreover, it is also noticed that the accuracy of cure depth predictions using this model can be as low as 50 % [3]. This issue compelled researchers to find a more sophisticated cure model capable of more accurate cure profile predictions. Lee et al. [5] developed a model that predicts the cure depth based on the given laser and resin parameters. It does so by modeling the chemical kinetics of the curing process and finding a quantitative relationship between the kinetics parameters and the cure depth.

Over the past two decades, multiple variations and extensions of the two models discussed above have been

developed to improve the accuracy of the cure profile predicted by these models. Tang et al. [3] presented a novel approach for modeling the curing process by accounting for laser exposure, photoinitiation, polymer chain propagation and termination, species diffusion in the curing polymer network, and heat transfer via conduction in the exposed region and its vicinity. A crucial distinction was made by considering the gel point to correspond to a threshold value of the degree of conversion (DOC) rather than a threshold on the exposure energy. This led to a much more accurate prediction of the cure profile than using a threshold on energy exposure. Jariwala et al. [4] proposed a model that incorporated the effect of oxygen inhibition and diffusion on photopolymerization and presented convincing results of improved accuracy of cure depth prediction for acrylate resins. Photopolymerization cure models also have applications in other unconventional polymer-based additive manufacturing processes such as Volumetric Additive Manufacturing (VAM) [6]. Salajeghe et al. [7] presented the first numerical model for the VAM method and used this model to investigate the influence of key resin parameters on the sedimentation rate of the curing part.

Besides the curing models, several studies in the literature have attempted to optimize the recoating process. Pham and Ji [8] presented a detailed study of the feasibility of building parts at higher speeds than generally recommended by SLA manufacturers. They achieved these higher print speeds using increased blade sweep speeds and found several other advantages, including reduced mounding, sweeping, and wait times. Tan and Gibson [9] presented a numerical study on the recoating process in micro-stereolithography. In particular, they studied the variation of the resin's settling time on the resin's viscosity and the object/container size ratio. The study deduced that low viscosity resins and a small object/vat size ratio could shorten the settling time and lead to faster printing speeds. There have also been efforts to improve the standard blade-based recoating method for specific printing requirements, such as printing over a cavity [10]. Spangenberg et al. [11] studied the flow characteristics of the liquid photopolymer resin in the SL process and estimated a prediction for the orientation of the short fibers added to the resin. Such predictions can be used to optimize the printer to prevent any misalignment of the fibers in the printed part. Wang et al. [12] performed an experimental and numerical analysis on the leading-edge bulge effect during the recoating process in ceramic stereolithography. The leading-edge bulge effect refers to the slurry surface bulge at the leading edge of the solidified part during the recoating process in stereolithography (SL). This effect leads to dimensional errors in the products and even a possible collision between the blade and the built part. This is an issue that is inherent to blade-based recoating mechanisms. Hence, there has been a need to explore other

recoating mechanisms which do not suffer from this problem. Hafkamp et al. [13] conducted a very detailed trade-off analysis of various recoating systems presently used for vat polymerization. Such studies can be used to select the optimal recoating method for the given printing requirements. New recoating mechanisms are also being developed to overcome the limitations of the available ones. Camargo et al. [14] presented a simple and innovative recoating mechanism using two blades. This mechanism had the advantage of improved printing speed and economy.

Over the last two decades, researchers have made considerable advances in optimizing the resin and process parameters to improve the printing speed. While there are quite a few studies on blade-based recoating techniques, there aren't any comprehensive studies on bladeless recoating techniques. These techniques do not have limitations, such as the leading-edge bulge, and have the potential to manufacture better quality parts. There are also no comprehensive SLA process simulation models available for testing and optimizing the process parameters of an SLA printer. This paper seeks to fill these gaps by comprehensively studying the curing and recoating processes and establishing essential conclusions that can help optimize the printer's resin and process parameters.

This work aims to develop a digital model that simulates the working of an SLA 3D printer, considering curing and recoating processes. Section 2.1 presents a cure model to capture the photopolymerization of the resin. The model is solved numerically, and a parametric analysis of the cure profile on various resin and process parameters is conducted. Section 2.2 presents a model of a gravity-based recoating technique. A 2D model is developed using the CFD software OpenFOAM® to study the dynamic motion of the resin in response to the motion of the build platform. Simulations are carried out where various parameters are varied to analyze the effect of these parameters on the settling time. Following this, the cure and resin flow models are combined and analyzed. SLA printer manufacturers could use the inferences gained from these studies to maximize printing speeds.

2 Model description

As described earlier, the SLA 3D printing process can be categorized to occur in two stages: part curing and resin recoating. This section presents the modeling approaches involved in each of the stages. The curing stage is described using a photo-polymerization model. This is followed by a description of the recoating modeling approach.

2.1 Photopolymerization physics

The first objective of the work was to study the dependence of the cure profile obtained in the SLA process on important laser and resin parameters used in the SLA 3D Printer. Figure 2 shows the schematic of the curing process. It depicts a laser beam being scanned over a vat of photosensitive resin at a constant velocity of V_s . The region described by $\{z: z \geq 0\}$ represents the vat of resin. The laser beam is assumed to have a Gaussian intensity profile, as shown in the figure. The cured part of the resin is also shown for representation. The cured region is characterized by the cure depth (C_d) and the line width (L_w).

The numerical model described here is an extension of the existing Lee Model [5] for cure profile estimation, which considers the laser intensity profile (I) to be represented by the fundamental transverse mode (TEM_{00}). The laser is being scanned along the x -axis with a constant velocity (V_s) starting from the origin of the global coordinate system. The distance traversed by the beam-center at a given time, t , can be computed as the product of the velocity (V_s) and time (t). Thus, the coordinates of the beam center as a function of time (t) can be represented as $(V_s t, 0)$ in the global coordinate frame.

When the laser interacts with the resin, the laser photons are absorbed by the photo-initiators leading to the formation of free-radicals. Due to this absorption, the laser beam is attenuated as it propagates inside the resin. The spatial intensity variation in the resin accounting for the absorption by the photo-initiators can be represented as [5]

$$I(x, y, z, t) = I_{xy}(10^{-\epsilon[PI]z}) \\ = I_0 e^{-2\frac{(x-V_s t)^2 + y^2}{w^2}} (10^{-\epsilon[PI]z}), \quad (1)$$

where $2w$ is the $1/e^2$ beam diameter, I_0 is the intensity at the centre of the beam, ϵ is the molar extinction coefficient ($M^{-1}l^{-1}$), and $[PI]$ is the molar concentration of the photoinitiator (M).

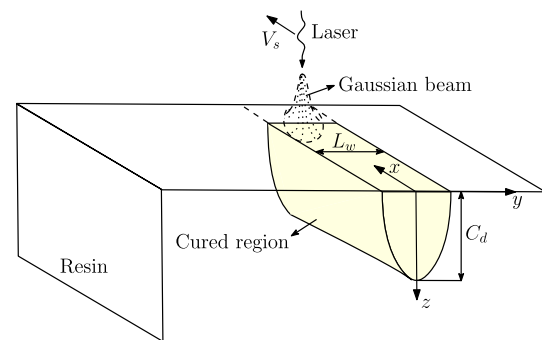


Fig. 2 Schematic of the curing process

The free-radicals formed due to laser absorption by photo-initiators leads polymerization of the (monomer) resin through a series of reactions [15]. The photoinitiators in the resin that have dissociated into free radicals attack the π -bonds of the monomers to form new radicals. This new radical attacks a monomer, and the process repeats to add monomer molecules successively and the reactive center propagates continuously. This results in extensive linking between many monomers to form a polymer. It is worth mentioning that the resin model considered here is basic and does not account for higher-order effects such as vitrification, autopolymerization, and oxygen inhibition [4]. Polymer growth is terminated at some point by the destruction of the reactive center by an appropriate reaction depending on the type of reactive center and the particular reaction conditions. This causes a decrease in monomer concentration. The rate of monomer disappearance is given by [5]:

$$-\frac{d[M]}{dt} - R_i = R_p = k_p[M] \left(\frac{\phi \epsilon [PI] I_{xy} (10^{-\epsilon [PI] z})}{k_t} \right)^{0.5}, \quad (2)$$

where $[M]$ is the monomer concentration at a depth z in the resin, R_i and R_p represent the rates of initiation and propagation, k_p is the kinetic rate constant for propagation, ϕ is the quantum yield of the photoinitiator, and k_t is the kinetic rate constant for termination.

In addition to describing the intensity profile more accurately, the model also relaxes the approximation that Lee et al. [5] had made by considering the monomer disappearance rate to be equal to the rate of propagation instead of the sum of both the rate of propagation and rate of initiation. The monomer concentration in the resin satisfies the differential Eq. 2. If the monomer concentration is known, the degree of polymerization ($\bar{x}_n$) [16] can be computed as

$$\bar{x}_n = \frac{[M]_0}{[M]}. \quad (3)$$

The extent of polymerization (p) [5] can be expressed in terms of the degree of polymerization as

$$p = \frac{\bar{x}_n - 1}{\bar{x}_n}. \quad (4)$$

This model assumes that the resin solidifies when the extent of polymerization at a point is greater than a threshold value. This threshold value is termed the critical extent of polymerization (p_c). Equation 4 can be used to locate the points in the resin that have an extent of polymerization greater than this threshold value ($p > p_c$) at a particular instant of time. These locations represent the cured parts of the resin. Using the extent of polymerization as a criterion for curing instead of exposure energy will lead to more accurate predictions of the cure profile [3]. Thus, the model can be used

to compute the cured and uncured parts of the resin at any point in time of the laser scanning process. However, Eq. 2 cannot be solved analytically to get a closed-form solution of $[M]$. Hence, a numerical differentiation procedure has to be used to solve the equation. This procedure is described in detail in Sect. 3.1.

2.2 Resin flow physics

The gravity-based recoating process studied in this paper consists of 2 major movements. At first, the build platform moves down and immerses the part deep into the resin. This allows the resin at the surface to fill the region initially occupied by the part. It then moves back up and leaves a small gap between the top surface of the part and the surface of the resin. This process is shown below in Fig. 3. The time taken for this process to take place is defined as the recoating time. It is a property of interest for manufacturers of printers, as it makes up for a significant percentage of the total build-time of a part.

The part being modeled here comprises stacked photopolymer “strings,” the fundamental building elements of stereolithography [2]. These stacked “strings” have a considerably large aspect ratio, so deflections in the resin fluid along the length of the strings would be minimal compared to those in the perpendicular plane. Hence, this 3-dimensional recoating process can be modeled in a 2D Cartesian domain. The domain is also symmetric about the central vertical line. Thus, the problem can be simplified further by modeling only one side of this axis of symmetry. Another point to note is that in practical scenarios, the build platform is usually far from the resin surface; hence, its effects on the surface motion are negligible. Consequently, it has been removed from the domain, and we study only the effect of the cured part moving through the resin. The final domain for the problem is represented in Fig. 4.

The flow of resin due to the motion of the build platform can be modeled as a laminar, incompressible flow. In addition, since the length scale of this problem is of the order of microns, continuum fluid mechanics are still valid, and hence the Navier–Stokes equations govern the motion of the resin. Inertial, viscous, pressure, gravitational, and surface tension forces drive this motion. These forces and the subsequent motion of the resin are represented by the 2D Navier–Stokes equations as

$$\nabla \cdot \vec{V} = 0 \quad (5)$$

and

$$\rho \frac{D\vec{V}}{Dt} = -\nabla p + \rho \vec{g} + \mu \nabla^2 \vec{V}. \quad (6)$$

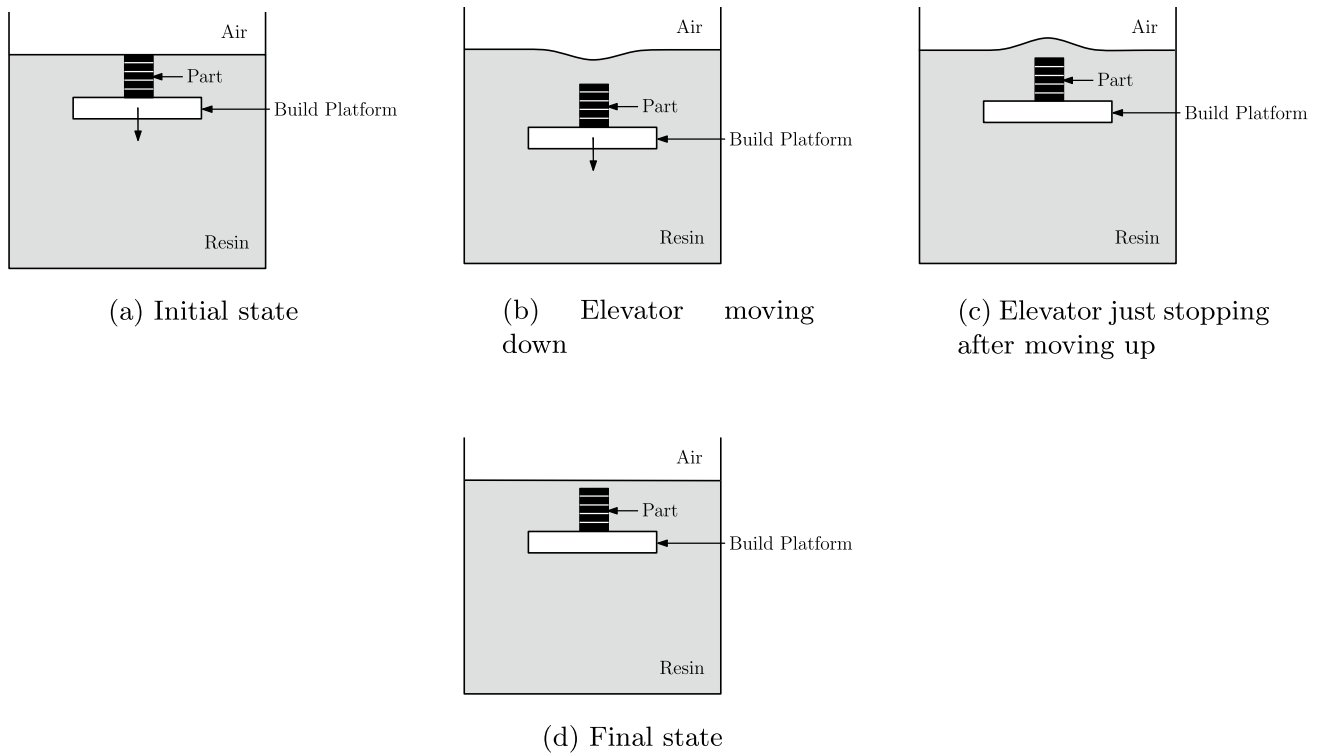


Fig. 3 Schematic of gravity-based recoating process

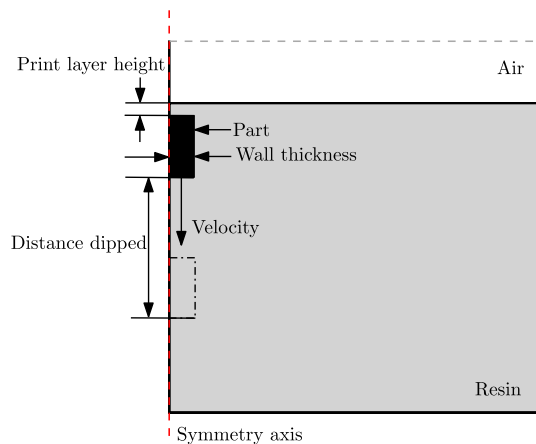


Fig. 4 Simplified model of the recoating process

The relative magnitude of the forces in play can be understood better by inspecting the dimensionless quantities that characterize the resin flow. The Reynolds number for this flow was computed to be 0.02, which is typical of a micro-scale flow. The Capillary number was found to be 0.24, indicating that both viscous and surface forces are of similar magnitudes. The Weber number for the case is just 0.004, which means that surface forces largely dominate the fluid inertial forces in the flow.

To track the surface profile of the resin, the motion of both the resin and air phases have to be tracked accurately. This is done by using a Volume of Fluid (VOF) model [17, 18] to simulate the multi-phase flow.

3 Computational model

3.1 Photopolymerization model

The monomer concentration at each point in the resin satisfies the first-order differential Eq. 2 in time. The Finite Difference Method (FDM) is used to solve Eq. 2 numerically. The local monomer concentration corresponding to the time t_i is represented as $[M]_i$. The first derivative of the monomer concentration is then expressed using the backward difference at time t_{n+1} . This changes Eq. 2 to

$$-\frac{[M]_{n+1} - [M]_n}{\Delta t} = k_p [M]_{n+1} \times \left(\frac{\phi \epsilon [PI](I_{xy}(x, y, t_{n+1}))(10^{-\epsilon [PI]z})}{k_t} \right)^{0.5} + 2\phi \epsilon [PI](I_{xy}(x, y, t_{n+1}))(10^{-\epsilon [PI]z}), \quad (7)$$

where Δt represents the time-step. The discretization error in the solution was minimized by ensuring that Δt was set to a value significantly smaller than the exposure time.

Hence, given the initial concentration of the monomer $[M]_0$, Eq. 7 is used to calculate the monomer concentration at time t_i by iteratively calculating the monomer concentrations at intermediate times ($t_1, t_2, \dots, t_{i-1}$). With the monomer concentration known, the extent of polymerization is computed using Eq. 4. The DOC threshold method is then used to compute the cure profile. The numerical solution method for the cure profile was implemented on Matlab R2021a by developing an in-house code. The solution was obtained in a 2D domain. The size of this domain was chosen such that it completely encompasses the cured regions in the resin. Figure 5 depicts a schematic of this discretized domain. The value of Δz here affects the accuracy of the solution. Smaller values provide a more accurate solution at the cost of increased computation time. Its value was chosen to minimize computation time while still ensuring that the solution accuracy is within permissible limits. Since cure depth is the most critical result, it was chosen as the parameter of interest in the mesh and time-step convergence study. The final change in the solution for the mesh convergence study was found to be 0.1%.

3.2 Resin flow model

Section 2.2 presented the 2D model of the resin flow problem. The governing equations of the problem are solved within this domain by employing CFD techniques. The numerical solutions to the governing equations of the problem are implemented on the open-source CFD software OpenFOAM® and are then post-processed in Paraview. The

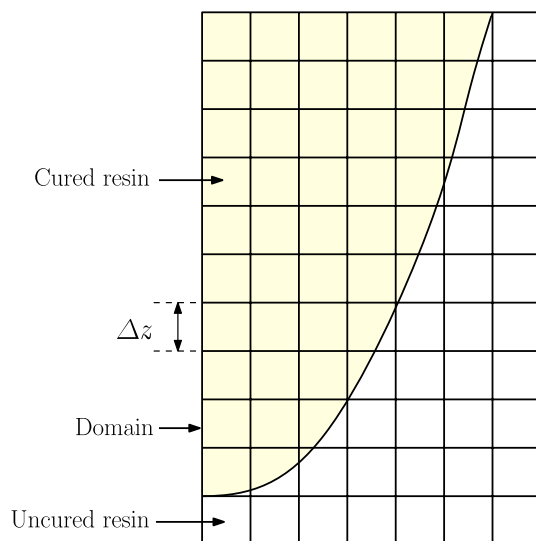


Fig. 5 Discretized domain

solver chosen in OpenFOAM®v2112 is interFOAM, which solves incompressible fluid flow problems and accounts for multiphase flow and surface tension effects. Table 1 presents the resin and process parameters used in the simulations [3, 19, 20].

Figure 6 presents the dimensions of the objects in the computational domain used for the simulations as well as the boundary conditions used in this problem. The bottom and right boundary conditions correspond to the walls of the container. The left wall boundary condition is a symmetry/zero gradient boundary condition to account for the symmetry of the problem solution. The top boundary condition allows both energy and mass transfer as the air above the resin can freely move in and out of this boundary. The solid body here represents the cured resin part. The boundaries of this solid body incorporate moving wall boundary conditions to effectively simulate the interaction of the fluid and the solid as the part moves through the resin.

Figure 7 depicts a picture of the mesh used. The mesh density is concentrated to the left of the resin, as the displacement of the resin is maximum at this region due to the moving platform and requires more cells to capture its motion accurately.

Table 1 Resin and recoating process parameters

Parameter	Symbol	Value	Units
Platform velocity	u	500	mm/min
Print layer height	d	0.2	mm
Distance dipped	D	2	mm
Resin density	ρ	1100	kg/m ³
Viscosity	μ	1	Pa.s
Surface tension	σ	0.035	N/m

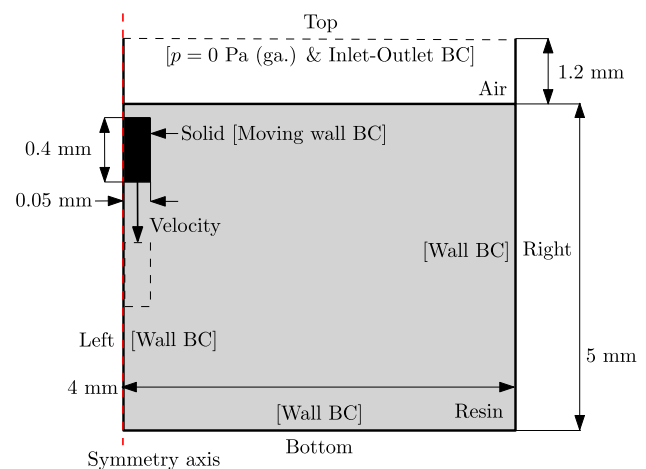


Fig. 6 Geometrical dimensions and Boundary conditions

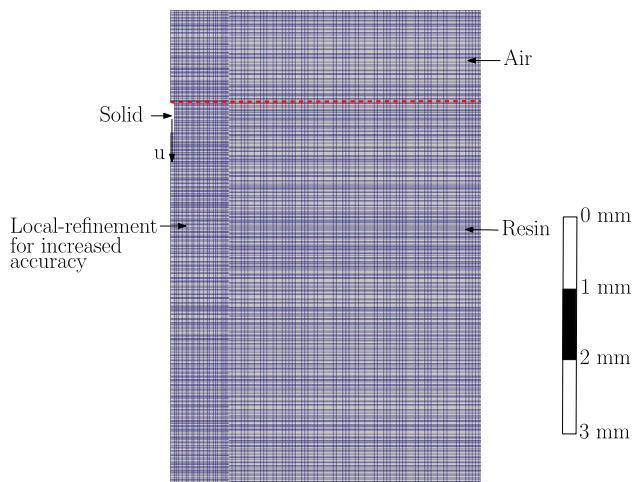


Fig. 7 2D mesh of the computational domain

Since settling time is one of the critical results of the simulations discussed below, it is crucial to define it precisely. In this paper, settling time is defined as the minimum time beyond which the deflections at the surface of the resin stay below a threshold value of 30 microns. To ensure the correctness of the numerical simulations, grid independence, and time-step independence studies were carried out to ensure that the final solution is mesh and time-step independent. Moreover, domain size independence studies were conducted to ensure that the computational size of the problem is as small as possible without compromising the accuracy of the solution.

4 Results and discussion

This section describes the results obtained from the photopolymerization model, recoating model, and the coupled model. The results obtained are also analyzed, and critical inferences are drawn from these studies.

4.1 Cure profile estimation

The results obtained by conducting simulations of the curing process are presented below. Various studies are conducted to observe the influence of key processing parameters on the cure profile of the resin.

The laser mimicked for this numerical study was a monochromatic laser of wavelength 325 nm. The laser beam had a Gaussian beam radius (w) of 110 μm , and its power was set to 28.8 mW. Table 2 contains important data about the resin used for conducting the studies. This resin is identical to the one used by Tang et al. [3].

Simulations were carried out to study the cure profile at the simulated plane as the laser is scanned over it at a

Table 2 Material parameter values of the resin

Parameter	Symbol	Value	Units
Monomer Concentration	$[M]$	2.136	mol/dm^3
Photoinitiator Concentration	$[PI]$	0.0898	mol/dm^3
Absorptivity (Initiator)	ϵ	19900	$\text{mol}^{-1}\text{dm}^3\text{m}^{-1}$
Initiation Quantum Yield	ϕ	0.6	(Unitless)
Rate Constant of Propagation	k_p	15.83	$\text{m}^3\text{mol}^{-1}\text{s}^{-1}$
Rate Constant of Termination	k_t	4190.3	$\text{m}^3\text{mol}^{-1}\text{s}^{-1}$
Critical Extent of Polymerization	p_c	0.09	(Unitless)

constant velocity of 2.72 cm/s in the x-direction. Figure 8 depicts the state of the cure at the simulated plane at various instances of time. The profile obtained is parabolic in nature, as expected.

Figure 9 presents the evolution of the cure depth and width as a function of the total exposure time for three different scanning velocities. One interesting observation is that although the Gaussian laser beam has an infinite extent, the rate of increase of the cure depth is not continuous. It suddenly spikes when the exposure time has reached a particular value and slowly decays to 0. A similar observation is made for the evolution of the cure width as well. This indicates that the DOC had to reach a particular value before the resin could be cured.

Figure 9 can also be used to compare the evolution of the cure profiles for the three different scanning velocities. It is observed that higher scanning velocities result in lower exposure times and consequently lower cure depths and cure widths as well. Also, while the difference in cure depths is quite significant, the cure width seems to be less sensitive to changes in scanning velocity.

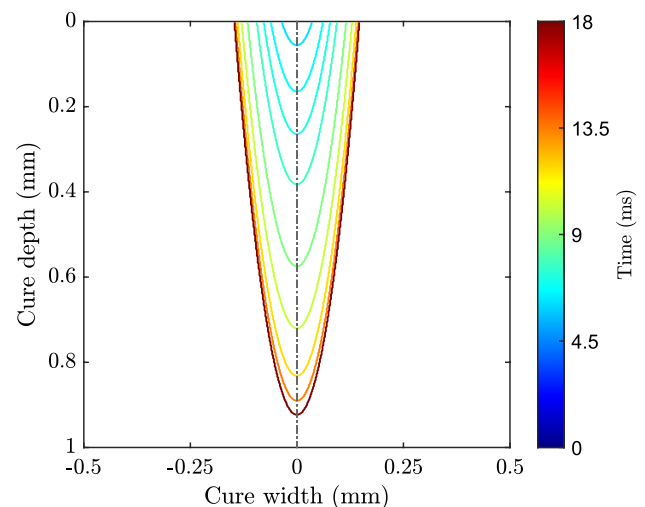
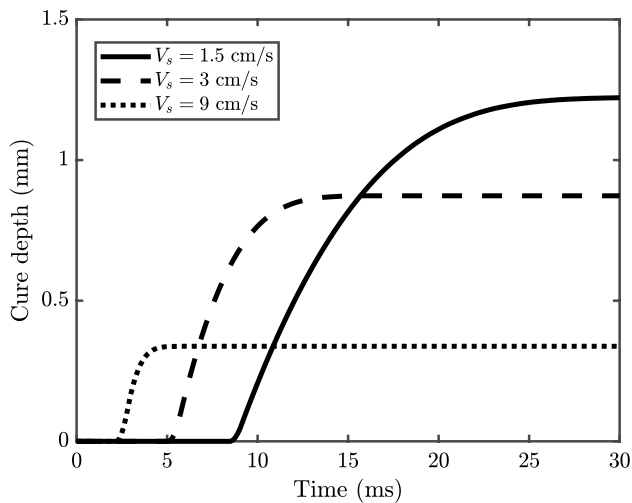
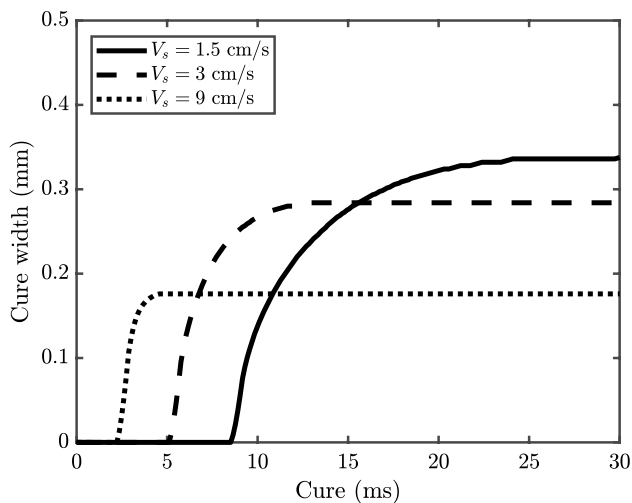


Fig. 8 Evolution of the cure profile over time



(a) Cure depth vs time



(b) Cure width vs time

Fig. 9 Comparison of the evolution of the cure profile over time for three scanning velocities

Another observation that is made from the two plots is that the convexity of the cure width curve is significantly higher than that of the cure depth curve. This indicates that while the cure depth steadily increases initially and then steadily decreases, the cure width increases sharply initially, and the rate of increase significantly decreases following that.

4.1.1 Variation with photoinitiator concentration

The resin composition has a massive effect on the printing process, and hence it is crucial to ensure that its composition is optimized. This optimization would be a multi-objective

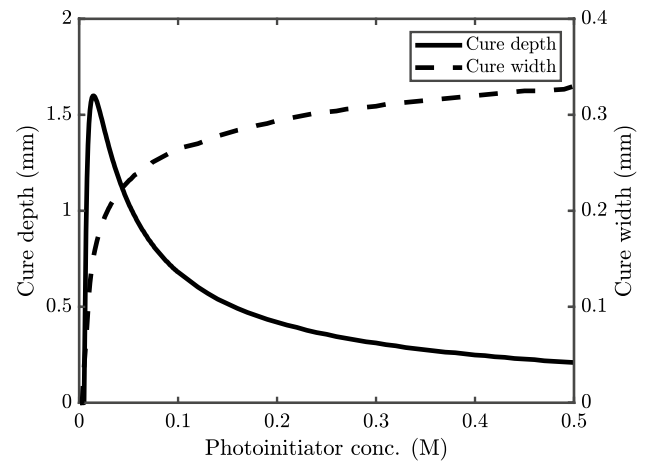


Fig. 10 Variation of the cure profile with change in photoinitiator concentration

optimization, keeping in mind various objectives such as maximizing the cure depth for a given scanning speed, maximizing the strength of the cured resin, and other such objectives. This study observed the effect of change in photoinitiator concentration on the cure profile. The study was conducted for a scanning speed of 4 cm/s and varied the photoinitiator concentration from 1 mM to 0.5 M. Figure 10 shows the variation of cure width and cure depth with change in photoinitiator concentration. As the photoinitiator concentration is increased from a minimal value, initially, there is a considerable rise in cure depth even for small changes in the photoinitiator concentration. It then reaches a maximum value for a particular value of photoinitiator concentration and then decreases continuously for further increase in the concentration. So clearly, there exists a value of photoinitiator concentration that maximizes the cure depth. Lee et al. [5] had made similar observations in his study of the variation of cure depth on photoinitiator concentration as well. On the other hand, it is seen that the cure width increases with photoinitiator concentration. It can also be noticed that the marginal increase in cure width is lower at higher photoinitiator concentrations.

4.1.2 Comparison of cure profile prediction models

Tables 3 and 4 show the predictions of the cure depth and width by various models [2, 3, 5] and the prediction error with respect to the experimental values. The experimental data here was obtained from the studies conducted by Tang et al. [3].

The proposed model could estimate the cure depth and width with good accuracy. It performed significantly better than the energy exposure threshold model. While the proposed model predicted the entire cure profile, the Lee Model could only predict the cure depth. The difference in

Table 3 Comparison of various cure depth prediction models

Model	Cure depth (μm)	Cure depth prediction error (%)
Experiment with 95% C.I	745 ± 10	–
Energy exposure threshold model	1118	50
Lee model	848	14
Tang model	660	– 10
Proposed model	920	23

Table 4 Comparison of various cure width prediction models

Model	Cure width (μm)	Cure width prediction error (%)
Experiment with 95% C.I	265 ± 4	–
Energy exposure threshold model	336	27
Lee model	–	–
Tang model	260	– 2
Proposed model	292	10

their predicted values for cure depth could be attributed to the difference in modeling of the laser intensity profile. Its deviation from the predictions of the Tang Model indicates that the effects of species diffusion and heat transfer in the polymer cure network are quite significant. Another point to note is that none of the models discussed here account for higher-order effects such as vitrification, autopolymerization, and oxygen inhibition [4]. Accounting for these effects would definitely help improve the cure prediction accuracy. However, modeling these effects requires accurate values of the resin's mechanical and chemical properties and involves higher model complexity. A multiphysics model of the SLA process could benefit greatly from the proposed cure model, which can give reasonably accurate results at low computational costs. It is also an especially good fit when the curing dynamics over time have to be captured. Hence, the proposed cure model would be a great choice in such cases where a computationally inexpensive model is required.

4.2 Recoating process

The recoating process, depicted earlier in Fig. 3, is simulated on OpenFOAM®, and the results of the study are presented in this section. The process involves the downward and upward movement of the part within the resin. The movement of the part attached to the build-platform is controlled such that it moves down and up at the same speed. Firstly, the cured part moves downward until it reaches a pre-decided depth. It then proceeds to move back up and stops,

leaving a certain gap between its top surface and the surface of the resin. This gap is meant for the next layer of resin to be cured using the laser. However, due to the momentum carried by the part on its way up, it causes the resin near it to move upwards as well and causes the surface to distort, as shown in Fig. 3c. Now, as time passes, the surface settles back into its flat shape and becomes ready for the next curing stage.

4.2.1 Evolution of resin shape

Figure 11 depicts the resin shape based on its phase diagram at a few critical time steps. Here, the time, t refers to the time elapsed after the build-platform has finished its motion. In Fig. 11a, it is noticed that the surface profile is significantly distorted immediately after the elevator has stopped moving. As explained in the previous section, this distortion was caused due to the upward momentum carried by the resin as the cured part moved through it. At $t = 2$ s, the deflections have reduced significantly, and by $t = 7$ s, it has been almost nullified. This indicates that the energy the system gained during the initial upward deflection of the resin surface gets damped out after a few oscillations. This energy loss can be attributed to viscosity and surface tension effects.

Figure 12 depicts the evolution of the maximum surface deflection over time. It can be seen that initially, there is a sharp rise in this deflection before it reaches its peak value and then proceeds to decrease sharply. As the resin surface falls due to gravity and surface tension effects, it picks up a downward momentum, and hence the surface deflects downward below the flat level. Once again, the surface and gravity forces drive it to rise back up to a second smaller peak. After this, the oscillations slowly die out until the surface is flat. Figure 12 also depicts a dashed line representing a threshold value of the surface deflection. This threshold is used to check whether the surface is suitable for the next curing stage to take place. The value of this threshold is chosen based on the required surface finish of the part. The surface is said to be “settled” when the deflections drop below this threshold and then stay below it until it dies down. The time it takes for this process to occur is known as the settling time. In Fig. 12, the settling time comes out to be around 10 s. Manufacturers of SLA 3D printers try to minimize this settling time to increase the overall print speed of the part. This study seeks to observe how this settling time varies with various resin parameters. The parameters considered in this study are the resin surface tension and the resin viscosity.

4.2.2 Parametric analysis of settling time

The surface tension of the resin can be controlled by adding various additives to the resin. Simulations were carried out

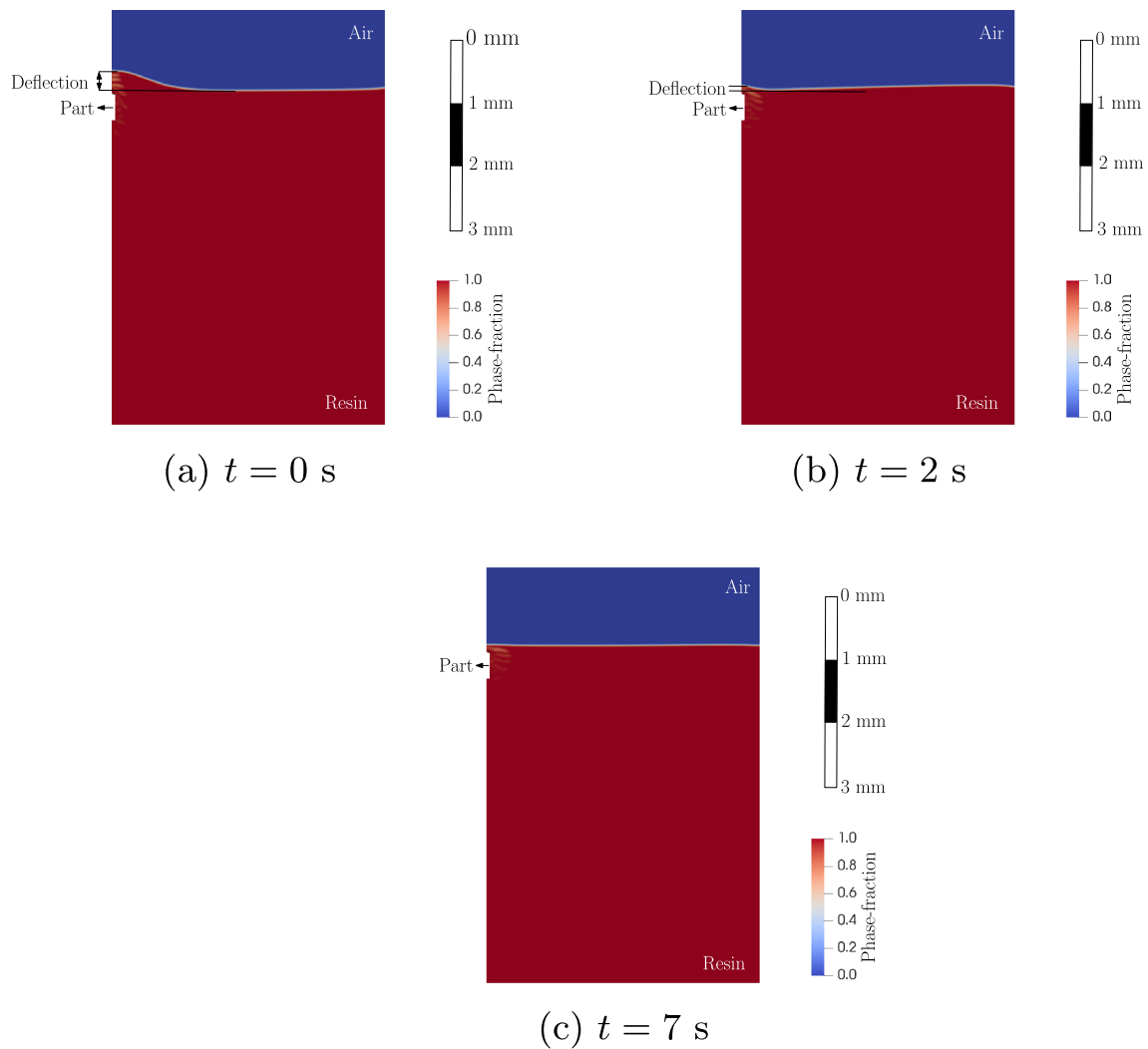


Fig. 11 Resin surface profile

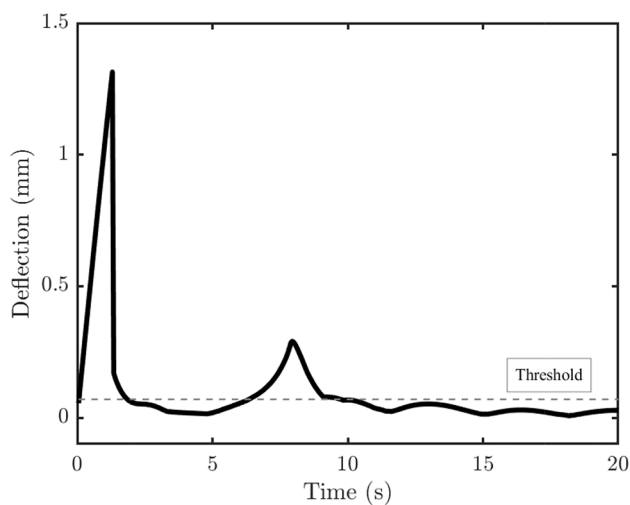


Fig. 12 Surface deflection vs time

to investigate its effect on settling time, and Fig. 13 depicts this variation. When increasing the surface tension by 120 %, it was observed that the settling time decreased by only 8 %. Hence, changing the surface tension of the resin does not seem to be a very efficient way to reduce the settling time.

Figure 13 also depicts the variation of settling time with resin viscosity. It is noticed that there is a significant increase in settling time with an increase in viscosity. This can be attributed to the fact that a more viscous resin will encounter more resistance when it flows, leading to longer settling times.

While some of these inferences would still hold for certain complex geometries as well, the resin flow can be significantly altered for parts with specific geometries. For example, if the part being built has enclosed spaces in it, surface forces could play a more prominent role and change the flow characteristics. A parametric analysis of parts having such geometries could reveal different optimal values of

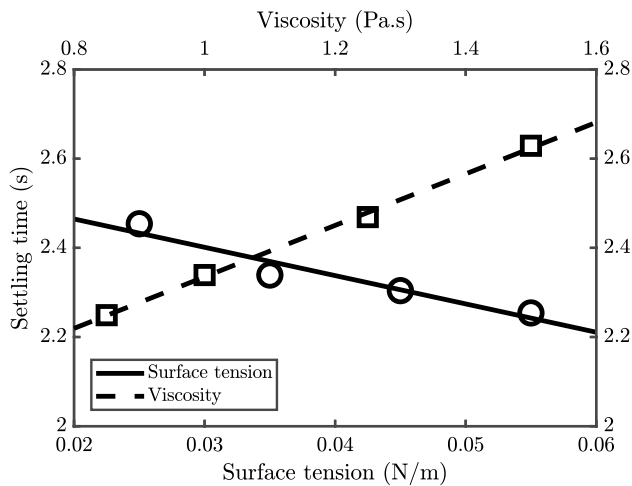


Fig. 13 Settling time vs surface tension and viscosity

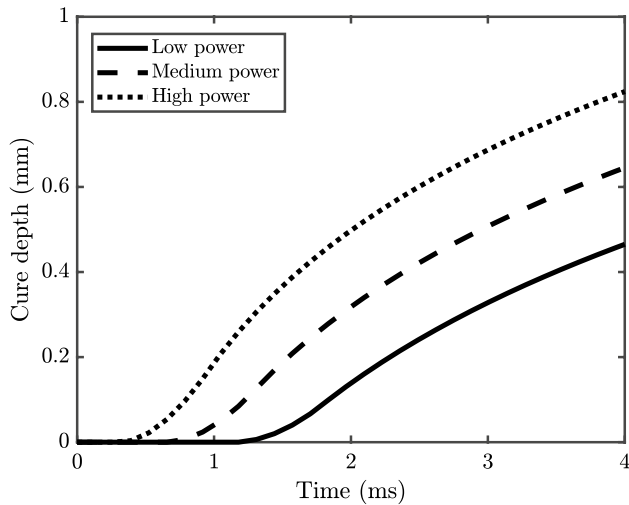
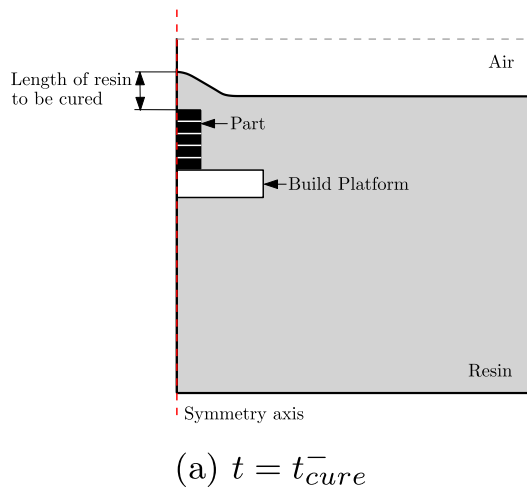


Fig. 14 Cure depth vs time



resin properties. This data could then be used to formulate resins that can be explicitly used to print parts having special geometries.

4.3 Coupled model

This section presents the studies conducted on the coupled model. Before the models could be coupled, a timescale analysis was performed to observe the closeness of the time-scales of the curing and recoating phenomena.

Figure 14 depicts the evolution of cure depth for 3 standard laser powers. On analysis of its growth over time, it turns out that the cure rate comes out to be $\sim 100 \mu\text{m/ms}$. Thus, it can be concluded that the time required for curing is of the order of 1 ms. Also, based on the resin flow studies presented in Sect. 2.2, the resin settling time is seen to be of the order of 1 s. Clearly, the resin flow motion takes three orders of magnitude more than the curing process. Thus, the process of photopolymerization of the resin can be treated as an instantaneous phenomenon with respect to the resin settling process. Thus, the problem of simultaneously simulating both the curing and recoating processes is now reduced to solving them one after the other. Firstly, the recoating process is simulated, and then, when the laser scanning starts, the resin is instantly cured. Following this, the recoating process is resumed. Figure 15 depicts a schematic of this concept. By implementing this concept alongside the equations governing the resin flow on OpenFOAM®, the simulations consider both phenomena, and these simulations were studied to gain insight into the printing process.

The above concept was implemented successfully in OpenFOAM® by using the software's topset, subsetmesh, and mapfields utilities. Figure 16 depicts the resin shape based on its phase diagram at a few critical time steps.

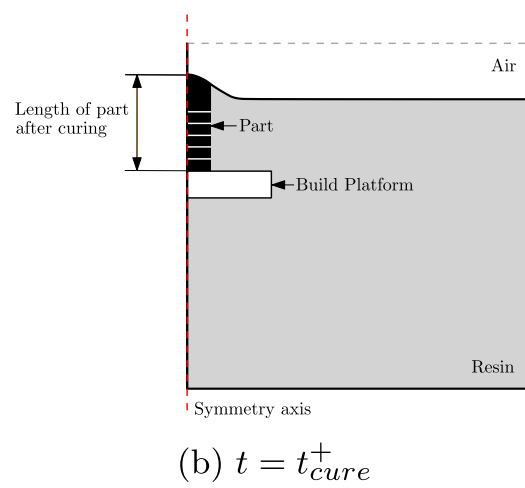


Fig. 15 State of cure before and after laser scanning

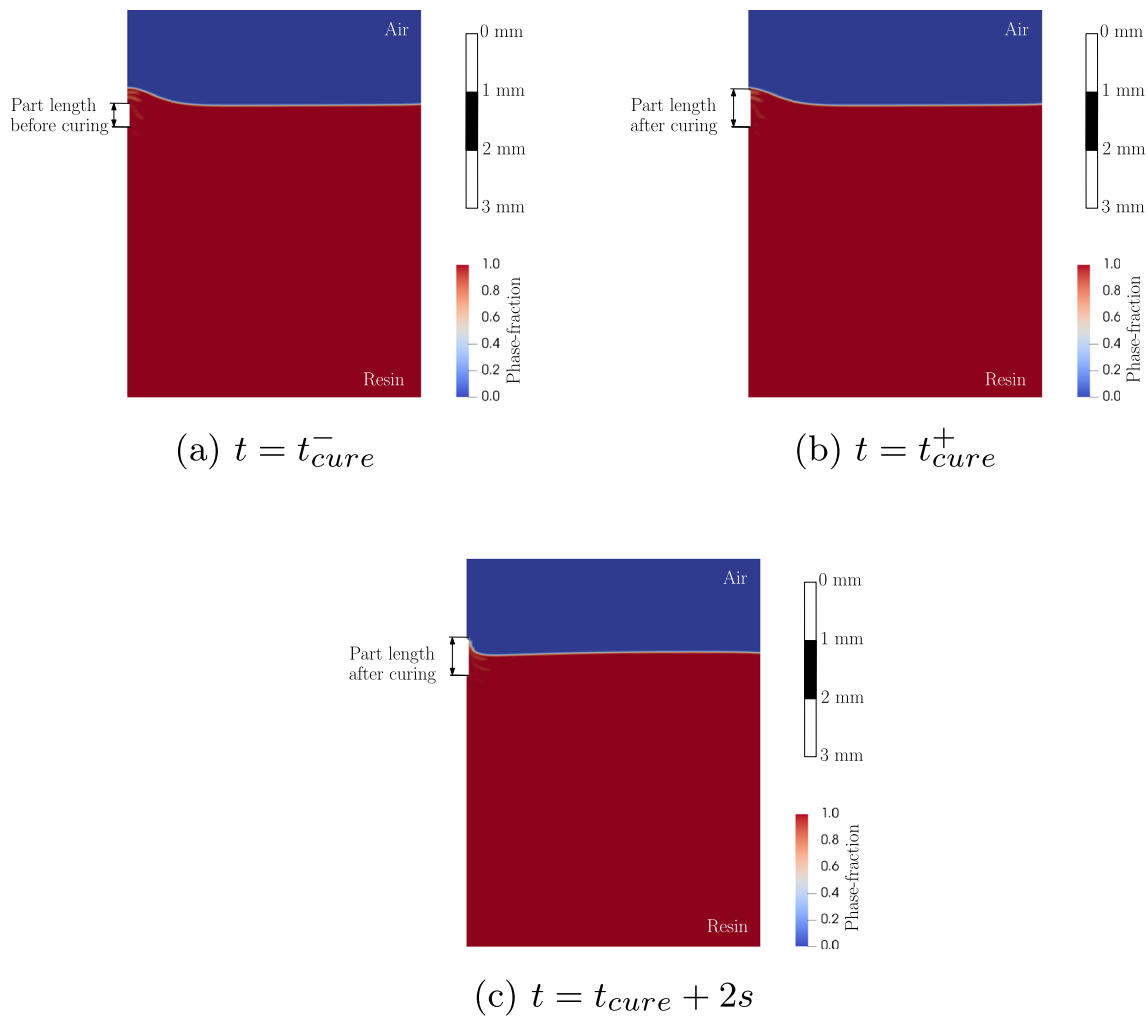


Fig. 16 Curing during the recoating process

The following sections present a couple of important studies conducted using this model. Section 4.3.1 describes how the surface profile of the newly cured layer can change as the recoating time is altered. Following this, Sect. 4.3.2 presents a brief study on how the surface quality of the printed part can be controlled by appropriately controlling the recoating time.

4.3.1 Variation of cure surface profile with recoating time

One method to increase the printing speed is to begin the curing process before the recoating process is complete. However, one disadvantage in doing so is that the surface profile of the cured region would not be flat. This could have dire consequences on the strength and surface characteristics of the final part. If the cured region is on the outer surface of the part, this will manifest itself in the form of irregularities in the surface of the finished part. On the other hand, if another “string” of resin is cured over this uneven

surface, it could lead to poor bonding between the two parts and thus lead to a potential source of failure in the part’s structural integrity. Data obtained from the simulations were post-processed to observe the potential cured surface profile that could be generated by curing the resin at different times. Figure 17 depicts these profiles when the resin is cured at different times after the elevator motion is complete.

On analyzing Fig. 17, it is noticed that the surface profile at $t = 0$ s has significant curvature, and this sort of surface cure profile could lead to multiple issues as described above. However, it can also be noticed that the profile quickly flattens out initially. Following this, it proceeds to settle very slowly. Hence, this intermittent period where most of the deflection has already been reduced could be a possible optimal time for curing the resin, as it would increase the printing speed without compromising too much on the build quality of the part.

Another aspect to consider here is that if there exists some deviation from flatness in the surface, it would also

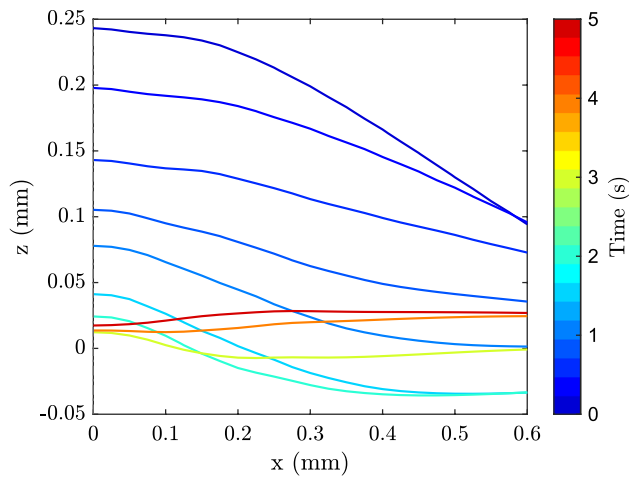


Fig. 17 Surface profile vs time

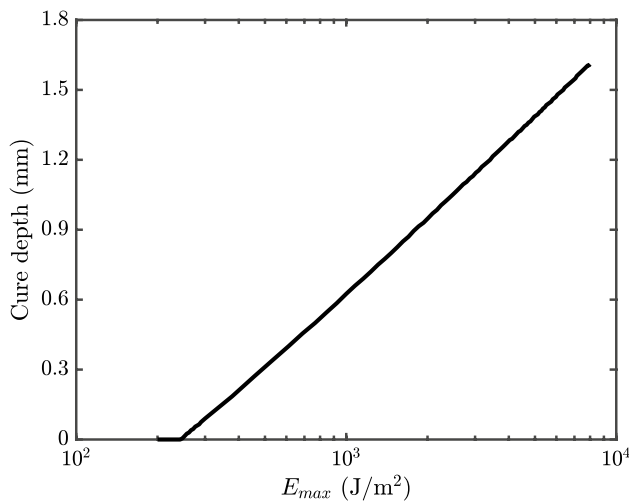


Fig. 18 Cure depth vs exposure energy

require a corresponding change in the laser power to account for the change in cure depth required. Figure 18 depicts the plot generated for this purpose. The exposure energy (E_{max}) refers to the maximum exposure of the laser at the surface of the resin. It is a function of the laser power (P) and the laser scanning speed (V_s) and is given by

$$E_{max} = \sqrt{\frac{2}{\pi}} \frac{P}{wV_s}. \quad (8)$$

Using Fig. 18 and Eq. 8, the laser power and the scanning speed required to cure a certain depth of resin can be easily computed. For example, to cure a resin up to a depth of 1 mm, the exposure energy required is 2244 J/m². A combination of realistic values of laser power and scanning speed

that could result in this exposure energy is a laser power of 28.8 mW and a scanning speed of 2.1 cm/s.

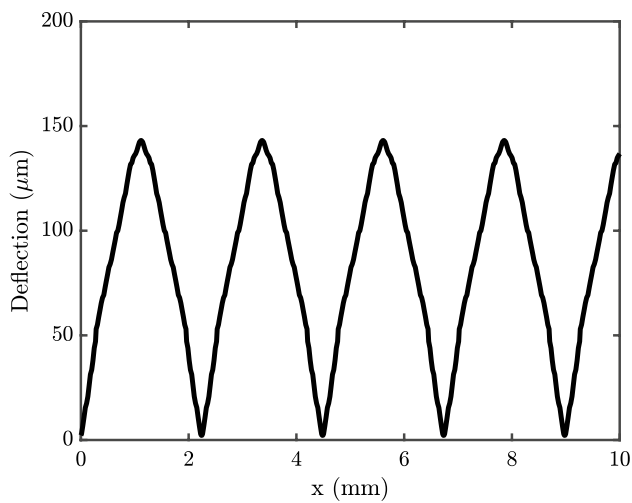
When analyzing Fig. 12 further, it is noticed that initially, there is a very sharp deflection at the surface. If the resin is cured within this period, the surface would be very uneven and would also require a higher laser power to cure it completely. As mentioned above, Fig. 18 can be referred to compute the increased laser power required for this. After the initial sharp deflection, it is noticed that there is a short period from approximately 2–6 s where the deflection remains below the threshold value before increasing again. Following the second deflection above the threshold, it remains below this value for the rest of the simulation. Similar observations are made on most of the simulations conducted. Thus, there is a potential opportunity to reduce the recoating time by choosing to cure the resin in the short window between the first and second deflection.

4.3.2 Effect of waiting time on surface quality

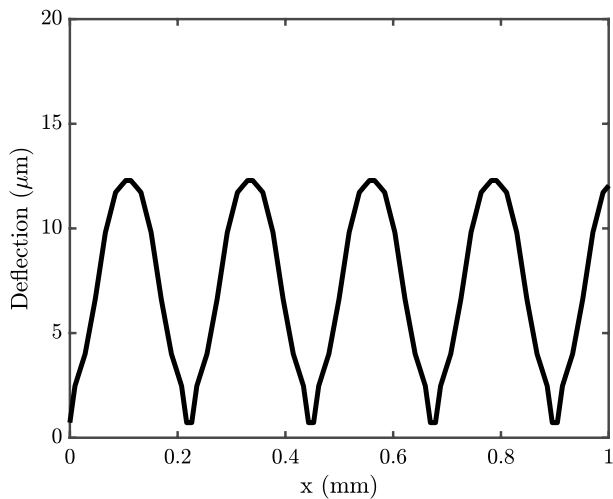
In this last study, the surface characteristics of a part are studied as the waiting time before it is cured, $t_{waiting}$, is varied. The part surface was generated as follows. The waiting time was fixed to a fairly low value, such that the resin had not entirely settled. At the time of cure, the laser is scanned such that all regions where the deflection is non-zero are cured. This results in a solidified protrusion being formed in the vat. It is assumed that if the entire process is repeated beside this protrusion, an identical protuberance will form next to it. Although this is a crude assumption, it is expected that it would work reasonably well for a preliminary analysis of the surface. This process is repeated beside each protrusion until the entire surface is cured. Such a surface would be rough with the presence of repeated peaks and valleys.

Figure 19 presents two such surface profiles generated using two different waiting times. Certain key roughness parameters for each one are also mentioned below the image. The scales of these two plots are also different. The deflections in the low waiting time case were significantly high, and hence the y-axis scale is one order of magnitude greater than that of the higher waiting time case. The arithmetic mean roughness (R_a) of the low waiting time case is consequently over ten times greater than that of the higher waiting time case. The kurtosis of roughness (R_{ku}) of both cases is lower than 3. This indicates that the surface is bumpy rather than spiky. Also, it is observed that R_{ku} of the higher waiting time case is slightly higher than that of the low waiting time case. Consequently, it can be seen that the high waiting time case has a more “spiky” surface as compared to the low waiting time case.

Thus, it is concluded that the appropriate choice of waiting time can significantly affect the surface characteristics



(a) $t_{\text{waiting}} = 0.5 \text{ s}$
 $R_a = 36 \text{ μm}$ & $R_{ku} = 1.9$



(b) $t_{\text{waiting}} = 3 \text{ s}$
 $R_a = 3.5 \text{ μm}$ & $R_{ku} = 1.6$

Fig. 19 Effect of waiting time on surface quality

of the printed part. While reducing the waiting time would increase the overall printing speed, it also leads to the poorer surface quality of the printed part. The above surface roughness study of the part can be repeated for a large set of waiting times, and a chart can be developed which maps a particular waiting time to the corresponding surface roughness characteristics. Given the surface roughness requirements of the part, the chart can be used to obtain the minimum waiting time, which would satisfy this requirement. This waiting time would ensure that the part is printed as fast as possible while still adhering to its surface quality requirements.

5 Conclusions

This paper presented a comprehensive study of the working of an SLA 3D printer. It first studied the photopolymerization physics involved in resin curing and then proceeded to study a gravity-based resin recoating mechanism for the printer. The two models were then coupled to understand how the two processes influence each other.

The photopolymerization model predicted the cure profile with decent accuracy while maintaining a low computational cost. It overestimated the cure depth by 23% and the cure width by 10%. Thus, the model is an ideal choice for first-order analysis and multiphysics models of the SLA process. It was found that the cure depth significantly decays with an increase in laser scanning speed. Also, a laser used at scanning speeds beyond a specific upper limit cannot cure the resin. This limit was found to be about 19 cm/s for the SLA setup considered in this paper. There also exists an optimum intermediate value of photoinitiator concentration that maximizes the cure depth. In contrast, it was found that the cure width increases continuously with photoinitiator concentration. The effects of the temporal variation of the laser intensity profile on the evolution of the cure profile were also studied. It was observed that the rate of growth of the cure profile during this process is high initially and slowly decays as the laser moves away. It should also be noted that this behavior would change if a different laser intensity profile is used instead.

The parametric study of the gravity-based recoating system revealed that resin viscosity is a very effective control parameter to alter the resin settling time. The settling time was found to vary linearly with the viscosity, and the sensitivity to changes in viscosity was found to be 0.58 s/Pa.s. Thus, a lower viscosity leads to a lower settling time. The study also revealed that the settling time could be reduced by increasing the resin surface tension. However, the settling time is significantly less sensitive to surface tension changes than viscosity changes. Hence, changing the surface tension of the resin is not a very efficient way to reduce the settling time.

Studies of the cure surface profile variation with recoating time found that there exists a short time frame between the first and second deflection spikes, where curing could potentially take place and reduce the recoating time by over 50 % without compromising too much on the part's quality. Deflections in the surface profile of the resin during recoating can also be compensated by appropriately tuning the laser power. It was also found that part surface roughness can be altered significantly by changing the waiting time for curing. Studies showed that a 2.5 s increase in waiting time led to a 90 % decrease in surface roughness. SLA printer manufacturers could now use the

insights gained from these studies to aid in the parameter calibration process of the printer.

Data availability The data that support the findings of this study are available from the corresponding author (Deepak Marla), upon reasonable request.

Declarations

Conflict of interest The authors have no relevant financial or non-financial interests to disclose.

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